Your Paper Title Here

Your Name(s)

Date

Abstract

Write your abstract here.

Keywords

Keyword 1, *Keyword 2*, *Keyword 3*, *Keyword 4*

Introduction / Background

Provide an introduction with background information.

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

Detail the likelihood approach.

Likelihood Inference

Let T denote the number of infections observed in the vaccine group among the total $n = 170$ infections recorded in the clinical trial. We assume

$$T \sim \text{Binomial}(n, \pi),$$

where $\pi = P(\text{infection occurs in the vaccine group} \mid \text{infection})$. In the observed data, $t = 8$ infections occurred in the vaccine group and 162 occurred in the placebo group.

The parameter of scientific interest is the vaccine efficacy ψ, defined as

$$\psi = \frac{\pi_p - \pi_v}{\pi_p},$$

where π_v and π_p denote the probabilities of infection in the vaccine and placebo groups respectively. Under the binomial formulation of the trial data, this relationship can be written in terms of π as

$$\psi = \frac{1 - 2\pi}{1 - \pi}.$$

Likelihood Function

The likelihood function for π is

$$L(\pi) = P(T = t|\pi) = \binom{n}{t} \pi^t (1 - \pi)^{n-t}.$$

Taking the logarithm gives the log-likelihood function

$$\ell(\pi) = \log L(\pi) = t \log(\pi) + (n - t) \log(1 - \pi).$$

The maximum likelihood estimator (MLE) is obtained by maximizing the likelihood function. For the binomial model, the MLE has the closed form

$$\hat{\pi}_{MLE} = \frac{T}{n}.$$

Because maximum likelihood estimators are invariant under transformation, the MLE for vaccine efficacy is

$$\hat{\psi}_{MLE} = \frac{1 - 2\hat{\pi}}{1 - \hat{\pi}}.$$

Large Sample Inference

Under regularity conditions and for sufficiently large sample sizes, the MLE is approximately normally distributed:

$$\hat{\pi}_{MLE} \approx N\left(\pi, \frac{\pi(1 - \pi)}{n}\right).$$

This result allows us to construct an approximate $(1 - \alpha)$ confidence interval for π using

$$\hat{\pi} \pm z_{\alpha/2}\sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}.$$

The endpoints of this interval can then be transformed using the relationship between ψ and π to obtain a confidence interval for vaccine efficacy.

Likelihood Ratio Test

To assess whether the vaccine efficacy meets the regulatory benchmark, we test the hypothesis

$$H_0 : \psi = 0.3 \qquad \text{versus} \qquad H_1 : \psi \neq 0.3.$$

The likelihood ratio test statistic is

$$W = 2\left[\ell(\hat{\pi}) - \ell(\pi_0)\right],$$

where π_0 corresponds to the null value of ψ and is obtained from

$$\pi_0 = \frac{1-\psi_0}{2-\psi_0}.$$

Under the null hypothesis and standard regularity conditions, the test statistic is approximately distributed as

$$W \sim \chi^2_1.$$

The corresponding p-value is obtained using the chi-square distribution.

Bootstrap Confidence Interval

To quantify uncertainty without relying solely on asymptotic theory, we also construct a parametric bootstrap confidence interval for ψ. Bootstrap samples are generated according to

$$T^* \sim \text{Binomial}(n, \hat{\pi}),$$

and the corresponding estimator $\hat{\psi}^*$ is calculated for each bootstrap sample. The empirical 2.5% and 97.5% quantiles of the bootstrap distribution provide a 95% percentile confidence interval for ψ.

Bayesian Inference

Detail the Bayesian approach.

Results

Present your findings.

```
# Data from Pfizer trial
n <- 170        # total infections
t <- 8          # infections in vaccine group


# ------------------------
# MLE for pi
# ------------------------
pi_hat <- t/n
pi_hat
```

```
## [1] 0.04705882
```

```
# ------------------------
# MLE for vaccine efficacy psi
# psi = (1 - 2*pi)/(1 - pi)
# ------------------------
```

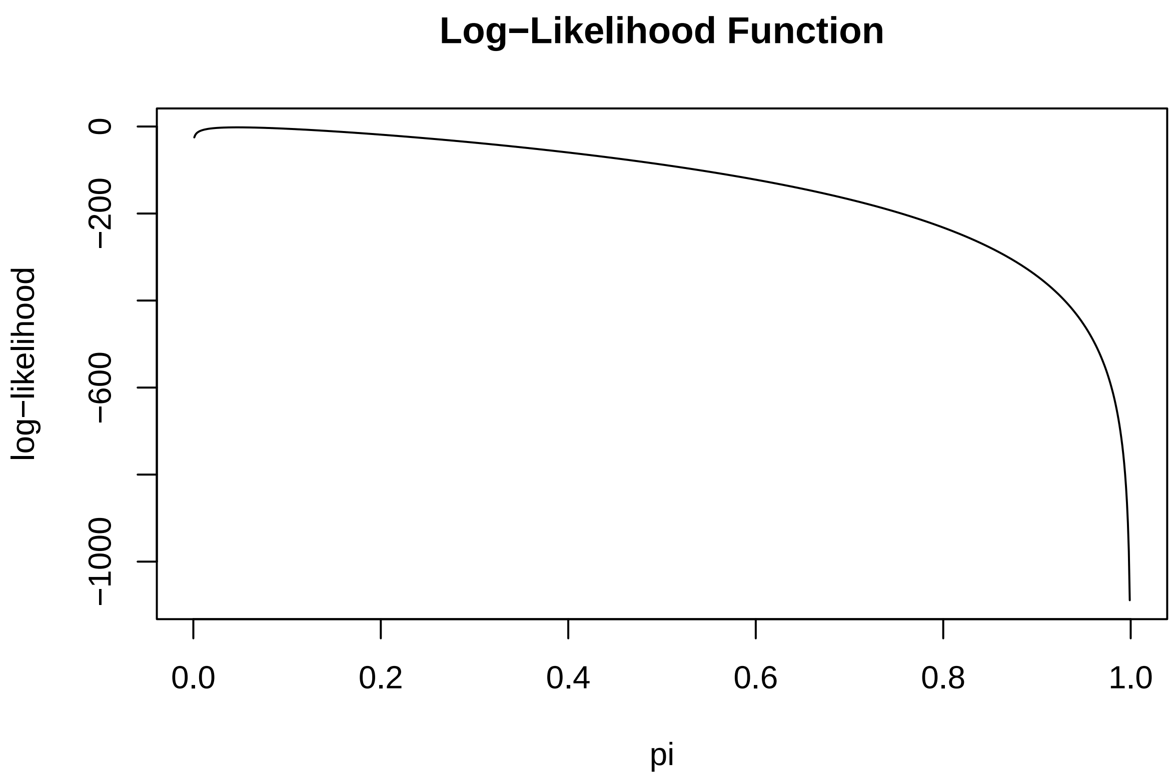
```r
psi_hat <- (1 - 2*pi_hat)/(1 - pi_hat)
psi_hat
```

```
## [1] 0.9506173
```

```r
# -------------------------
# Log-likelihood function
# -------------------------
loglik <- function(pi, t, n){
  if(pi <= 0 | pi >= 1) return(-Inf)
  dbinom(t, size=n, prob=pi, log=TRUE)
}

# evaluate likelihood across grid
pi_vals <- seq(0.001,0.999,length=1000)
ll_vals <- sapply(pi_vals, loglik, t=t, n=n)

plot(pi_vals, ll_vals, type="l",
     xlab="pi",
     ylab="log-likelihood",
     main="Log-Likelihood Function")
```




```r
# -------------------------
# Large sample CI for pi
# -------------------------
se_pi <- sqrt(pi_hat*(1-pi_hat)/n)


ci_pi <- pi_hat + c(-1,1)*qnorm(0.975)*se_pi
ci_pi
```

```
## [1] 0.01522585 0.07889180
```

```r
# -------------------------
# Convert CI to psi scale
# -------------------------
psi_transform <- function(pi){
  (1-2*pi)/(1-pi)
}
```

```r
ci_psi <- psi_transform(ci_pi)
ci_psi
```

```
## [1] 0.9845387 0.9143512
```

```r
# -------------------------
# Likelihood Ratio Test
# H0: psi = 0.3
# -------------------------

psi0 <- 0.3

# convert psi0 to pi0
pi0 <- (1 - psi0)/(2 - psi0)

# likelihood ratio statistic
W <- 2*(loglik(pi_hat,t,n) - loglik(pi0,t,n))
W
```

```
## [1] 121.6012
```

```r
# p-value from chi-square distribution
p_value <- 1 - pchisq(W, df=1)
p_value
```

```
## [1] 0
```

```r
# -------------------------
# Parametric bootstrap CI
# -------------------------
B <- 10000

boot_pi <- rbinom(B, n, pi_hat)/n
```

```r
boot_psi <- psi_transform(boot_pi)


boot_ci <- quantile(boot_psi, c(0.025,0.975))
boot_ci
```

```
##      2.5%     97.5%
## 0.9102564 0.9820359
```

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from https://www.example.com.

Appendix

Code

```r
# You can reference your code in the appendix (sample here).
```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.